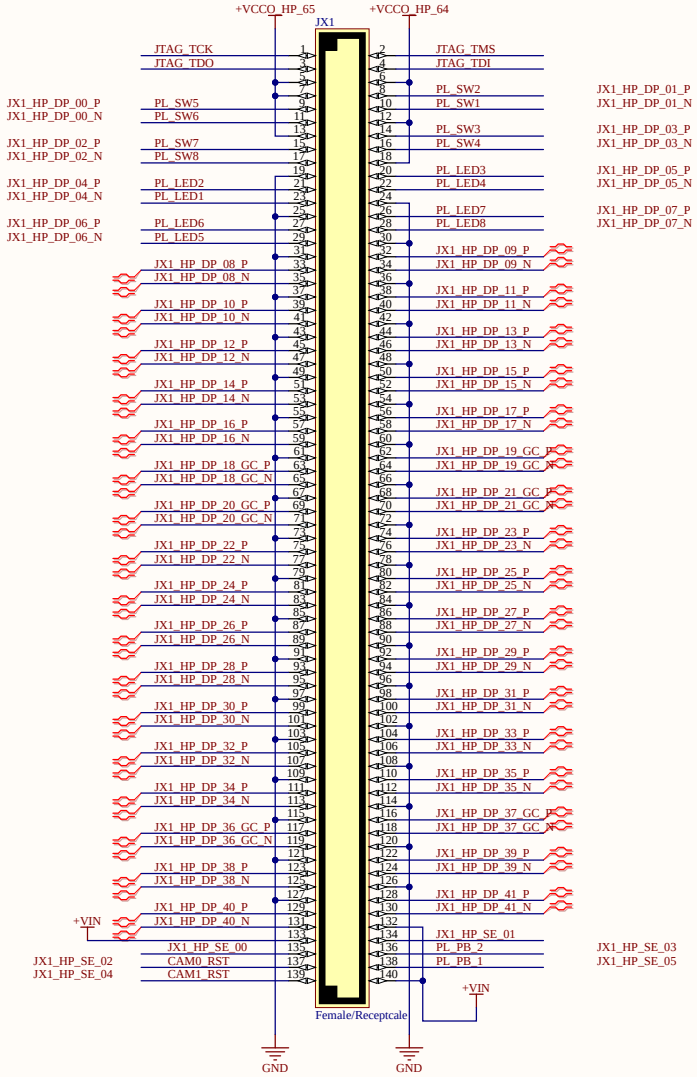
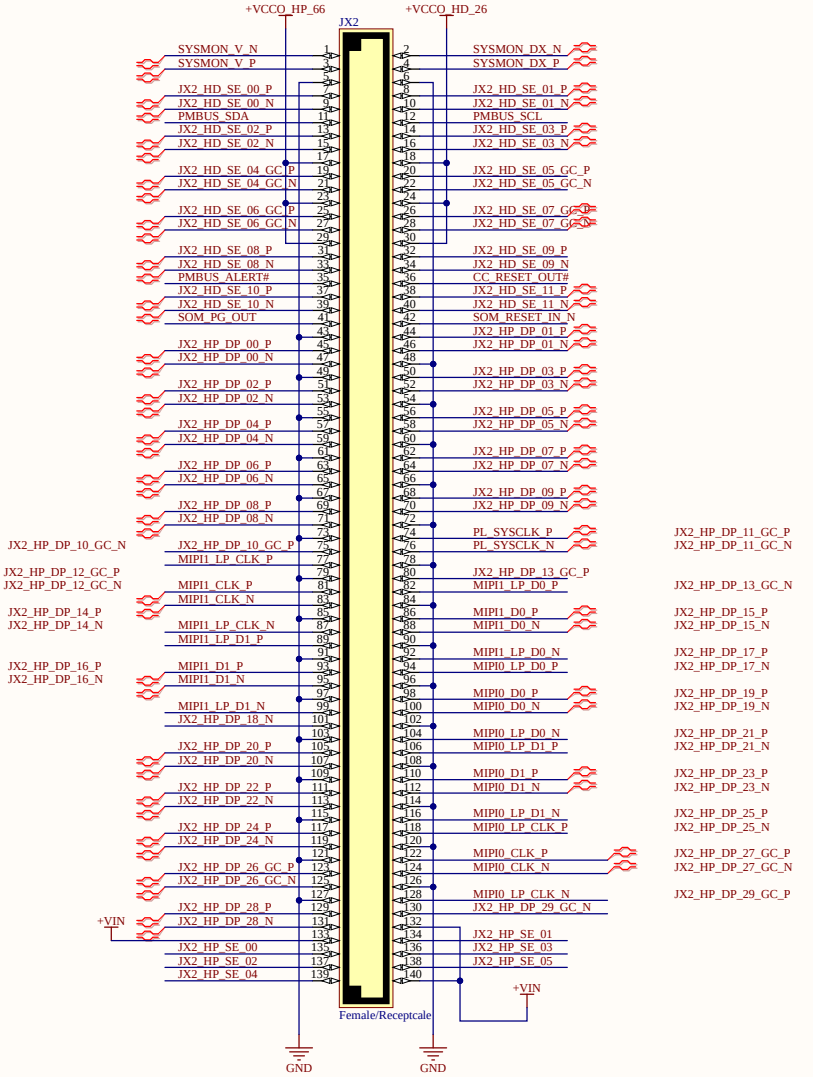
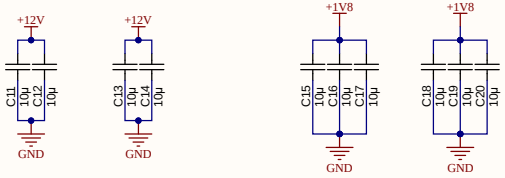
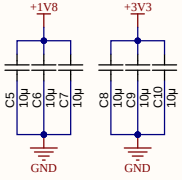
+VCCO_PSIO, 1V8, JTAG is +VCCO_PSIO

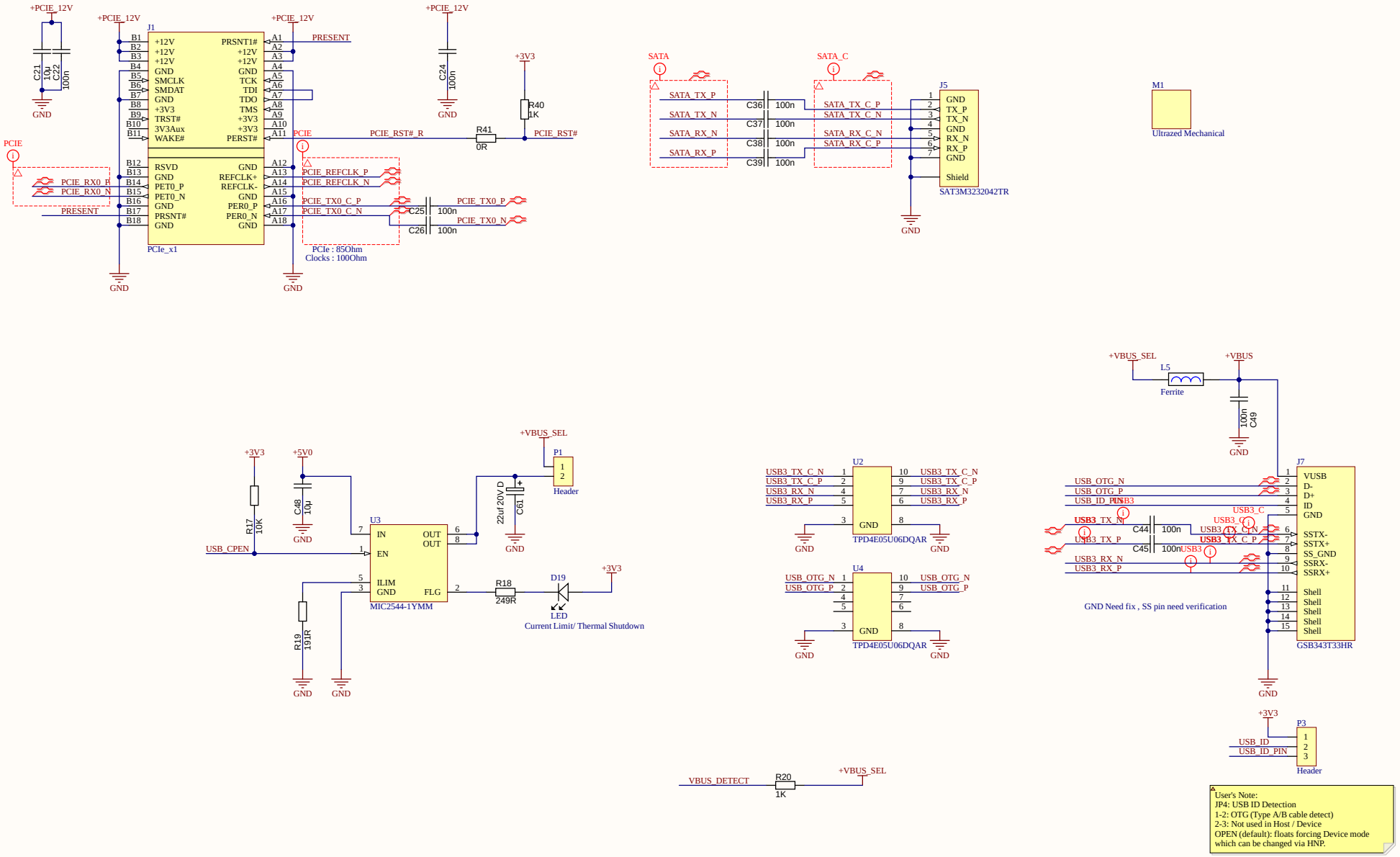


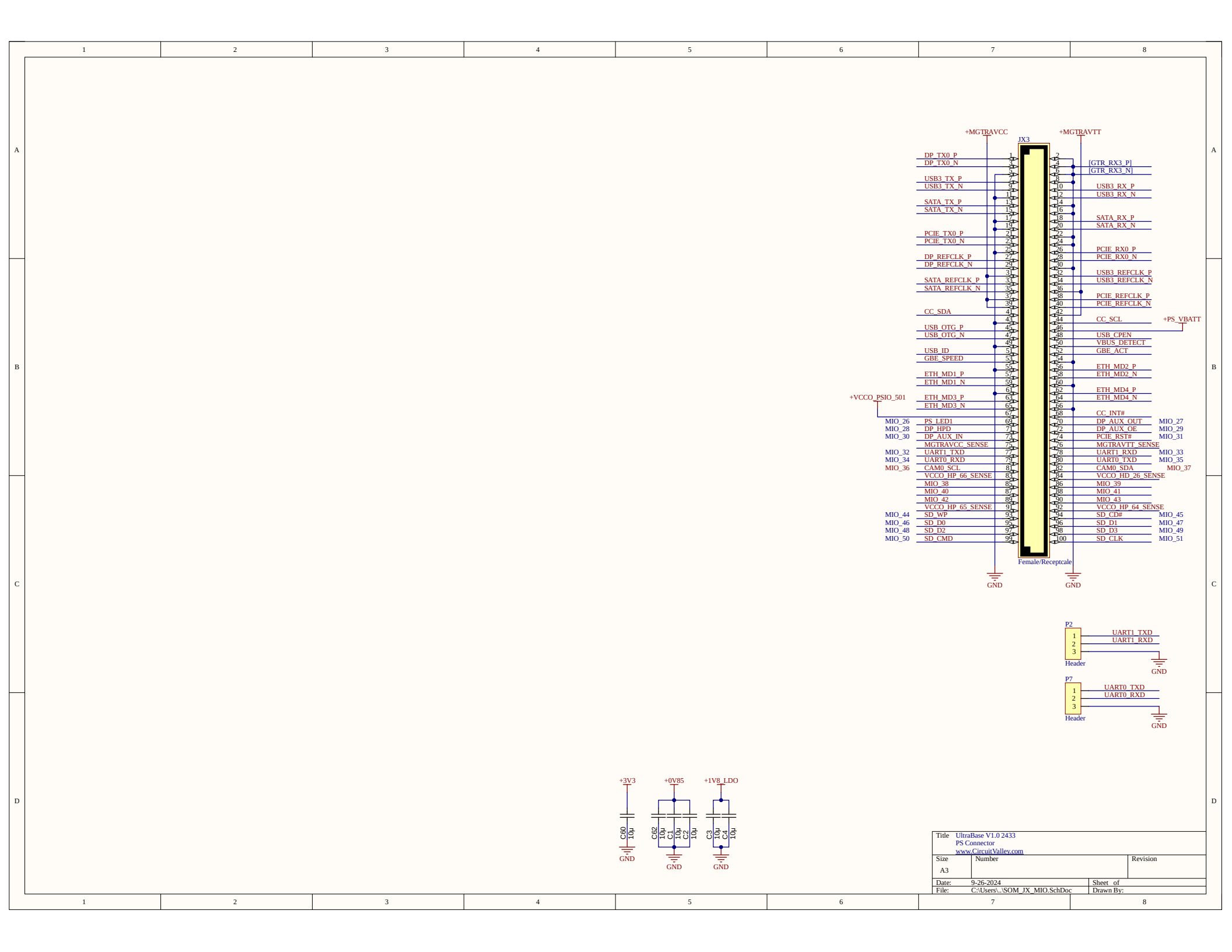




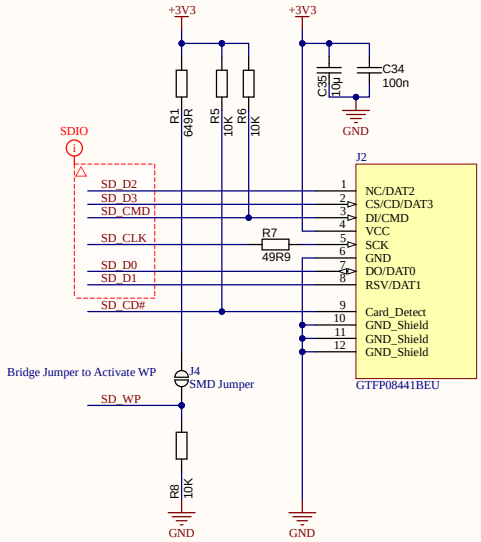


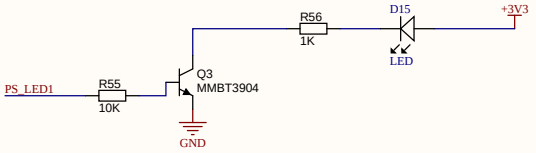
Title	UltraBase V1.0 2433 PL Connectors www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\PL_Conenctor.schDoc		Drawn By:


+PCIE_12V
C21 10µ
C22 100n
GND
J1
PCIe_x1
PCIE
PCIE_RX0_P
PCIE_RX0_N
PRESENT
PCIE_RST#_R
R41 0R
PCIE_RST#
PCIE_REFCLK_P
PCIE_REFCLK_N
PCIE_TX0_C_P
PCIE_TX0_C_N
C25 100n
C26 100n
PCIE_TX0_P
PCIE_TX0_N
PCIe : 85Ohm
Clocks : 100Ohm
C24 100n
+3V3
R40 1K
SATA
SATA_C
SATA_TX_P
SATA_TX_N
SATA_RX_N
SATA_RX_P
C36 100n
C37 100n
C38 100n
C39 100n
J5
SAT3M3232042TR
M1
Ultrazed Mechanical
+VBUS_SEL
P1
Header
+5V0
R17 10K
C48 10µ
U3
MIC2544-1YMM
USB_CPEN
22uf 20V D
C61
R18 249R
D19
LED
Current Limit/ Thermal Shutdown
R19 191R
U2
TPD4E05U06DQAR
U4
USB3_TX_C_N
USB3_TX_C_P
USB3_RX_N
USB3_RX_P
USB_OTG_N
USB_OTG_P
+VBUS
L5
Ferrite
C49 100n
J7
GSB343T33HR
USB_OTG_N
USB_OTG_P
USB_ID_PIN
USB3_TX_N
USB3_TX_P
USB3_RX_N
USB3_RX_P
C44 100n
C45 100n
GND Need fix , SS pin need verification
P3
Header
USB_ID
VBUS_DETECT
R20 1K
User's Note:
JP4: USB ID Detection
1-2: OTG (Type A/B cable detect)
2-3: Not used in Host / Device
OPEN (default): floats forcing Device mode
which can be changed via HNP.
Title UltraBase V1.0 2433
PCIe, SATA USB OTG
www.CircuitValley.com
Size A3
Number
Revision
Date: 9-26-2024
File: C:\Users\..\PCIe.SchDoc
Sheet of
Drawn By:

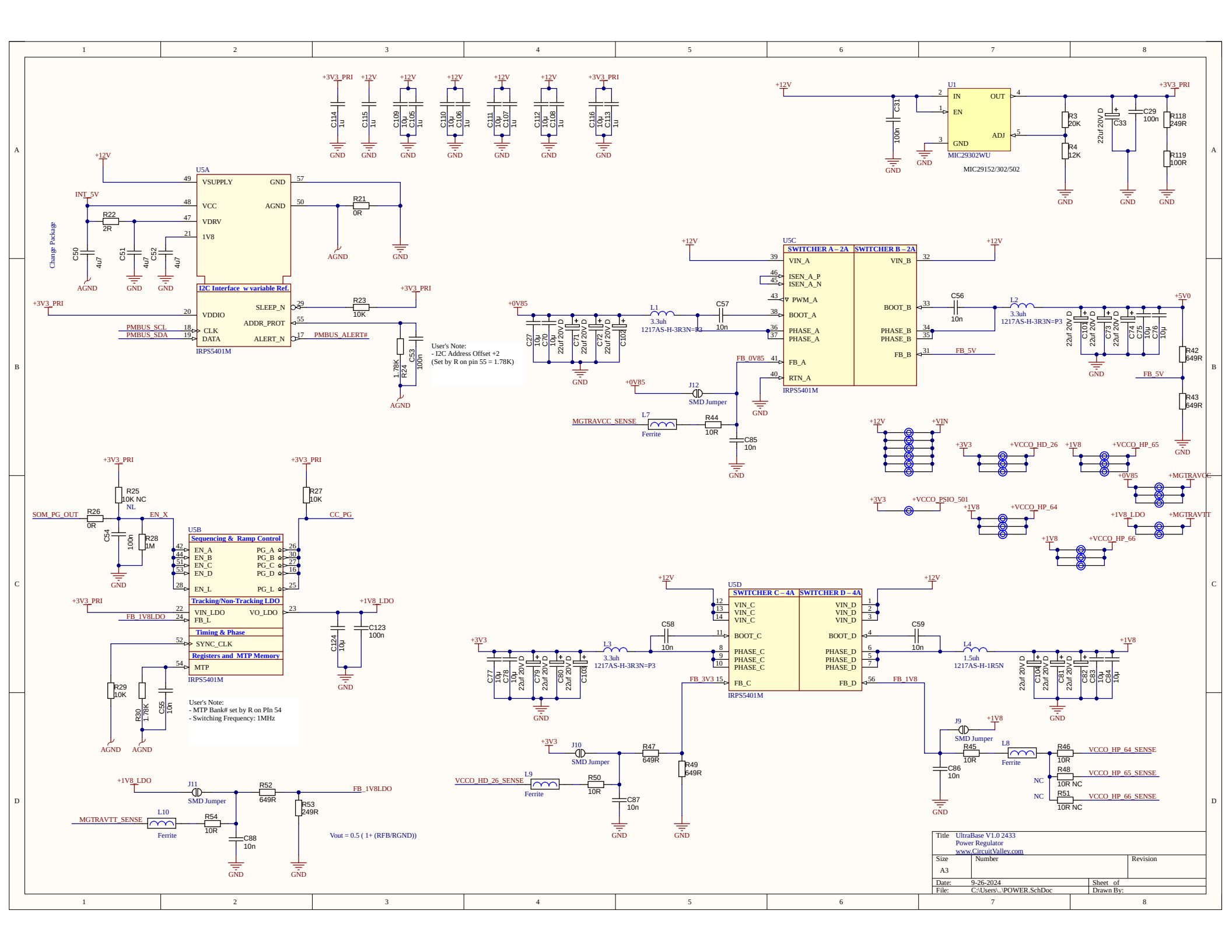


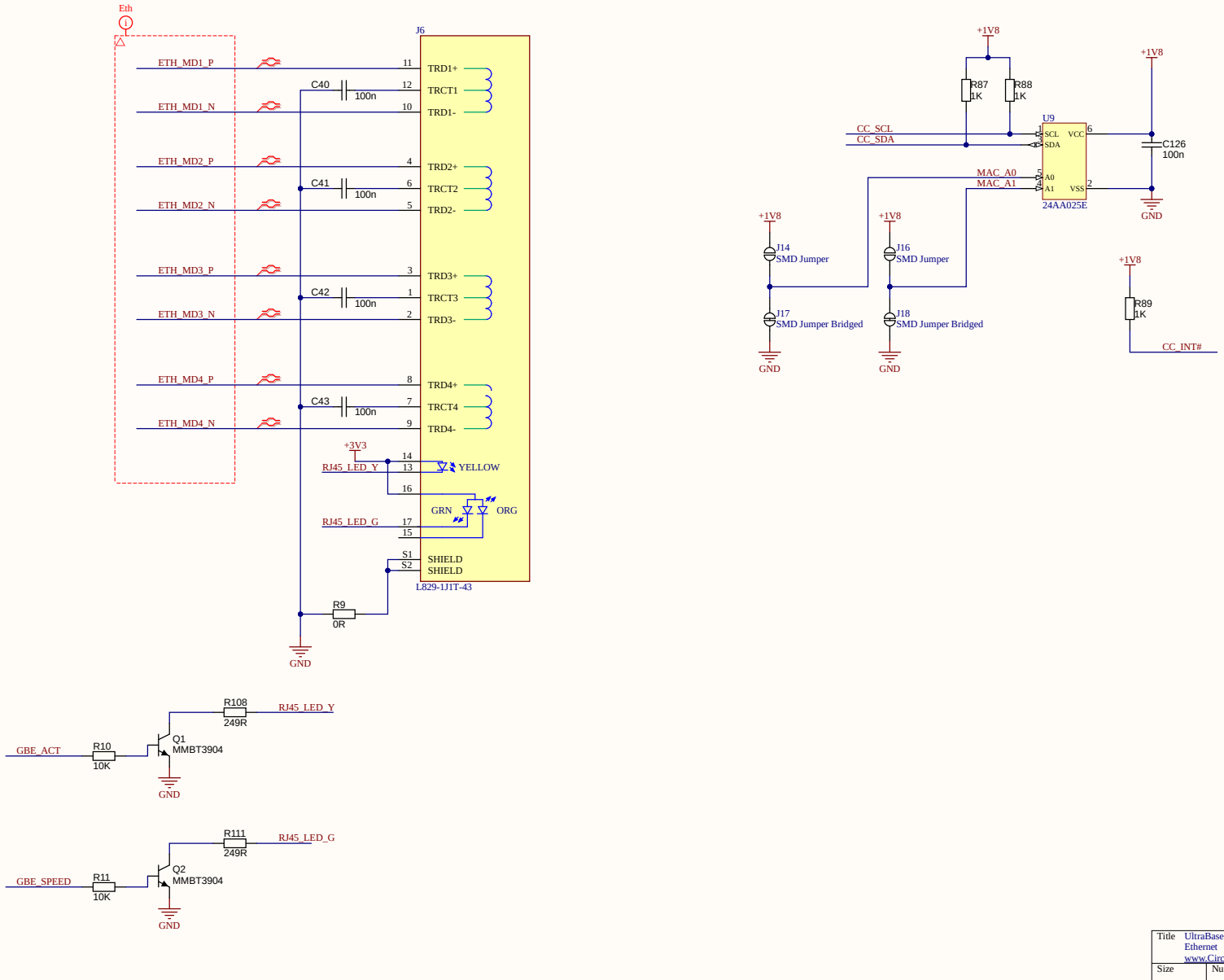
Title	UltraBase V1.0 2433 PS Connector www.CircuitValley.com	
Size: A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\SOM_JX_MIO.SchDoc		Drawn By:



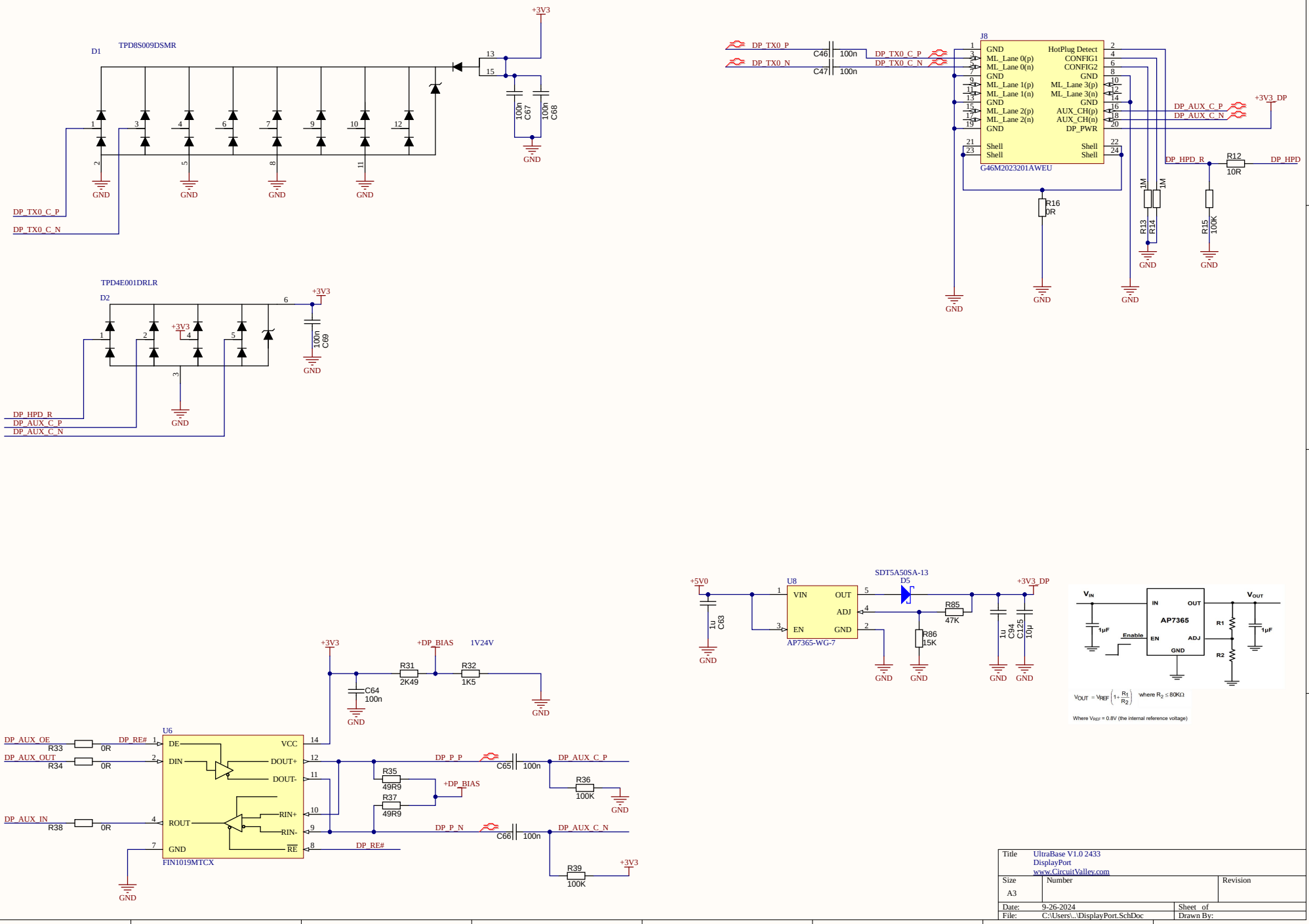


Title	UltraBase V1.0 2433 MIO SD www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\MIO_SD.SchDoc		Drawn By:

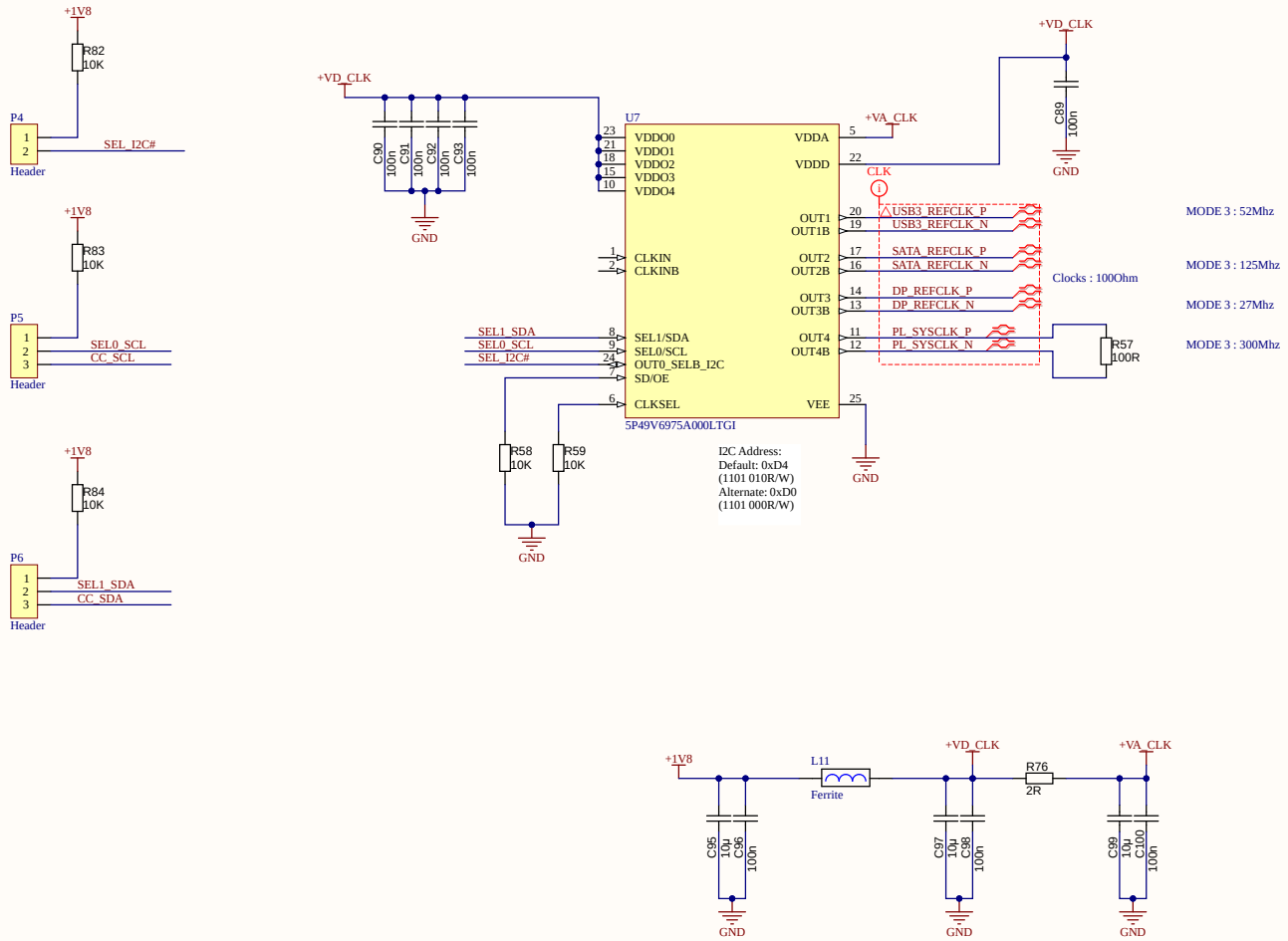

U5A
IRPS5401M
I2C Interface w variable Ref.
VSUPPLY
VCC
VDRV
1V8
VDDIO
CLK
DATA
GND
AGND
SLEEP_N
ADDR_PROT
ALERT_N
PMBUS_ALERT#
PMBUS_SCL
PMBUS_SDA
INT_5V
+3V3_PRI
+12V
Change Package
User's Note:
- I2C Address Offset +2
(Set by R on pin 55 = 1.78K)
U1
MIC29302WU
MIC29152/302/502
IN
OUT
EN
ADJ
U5C
SWITCHER A – 2A
SWITCHER B – 2A
VIN_A
VIN_B
ISEN_A_P
ISEN_A_N
PWM_A
BOOT_A
BOOT_B
PHASE_A
PHASE_B
FB_A
FB_B
RTN_A
+0V85
+5V0
FB_0V85
FB_5V
MGTRAVCC_SENSE
Ferrite
SMD Jumper
U5B
Sequencing & Ramp Control
EN_A
EN_B
EN_C
EN_D
EN_L
PG_A
PG_B
PG_C
PG_D
PG_L
Tracking/Non-Tracking LDO
VIN_LDO
FB_L
VO_LDO
Timing & Phase
SYNC_CLK
Registers and MTP Memory
MTP
SOM_PG_OUT
EN_X
CC_PG
FB_1V8LDO
+1V8_LDO
User's Note:
- MTP Bank# set by R on PIn 54
- Switching Frequency: 1MHz
U5D
SWITCHER C – 4A
SWITCHER D – 4A
VIN_C
VIN_D
BOOT_C
BOOT_D
PHASE_C
PHASE_D
FB_C
FB_D
FB_3V3
FB_1V8
+3V3
+1V8
VCCO_HD_26_SENSE
VCCO_HP_64_SENSE
VCCO_HP_65_SENSE
VCCO_HP_66_SENSE
NC
Vout = 0.5 (1+ (RFB/RGND))
Title UltraBase V1.0 2433
Power Regulator
www.CircuitValley.com
Size A3
Number
Revision
Date: 9-26-2024
Sheet of
File: C:\Users\..\POWER.SchDoc
Drawn By:



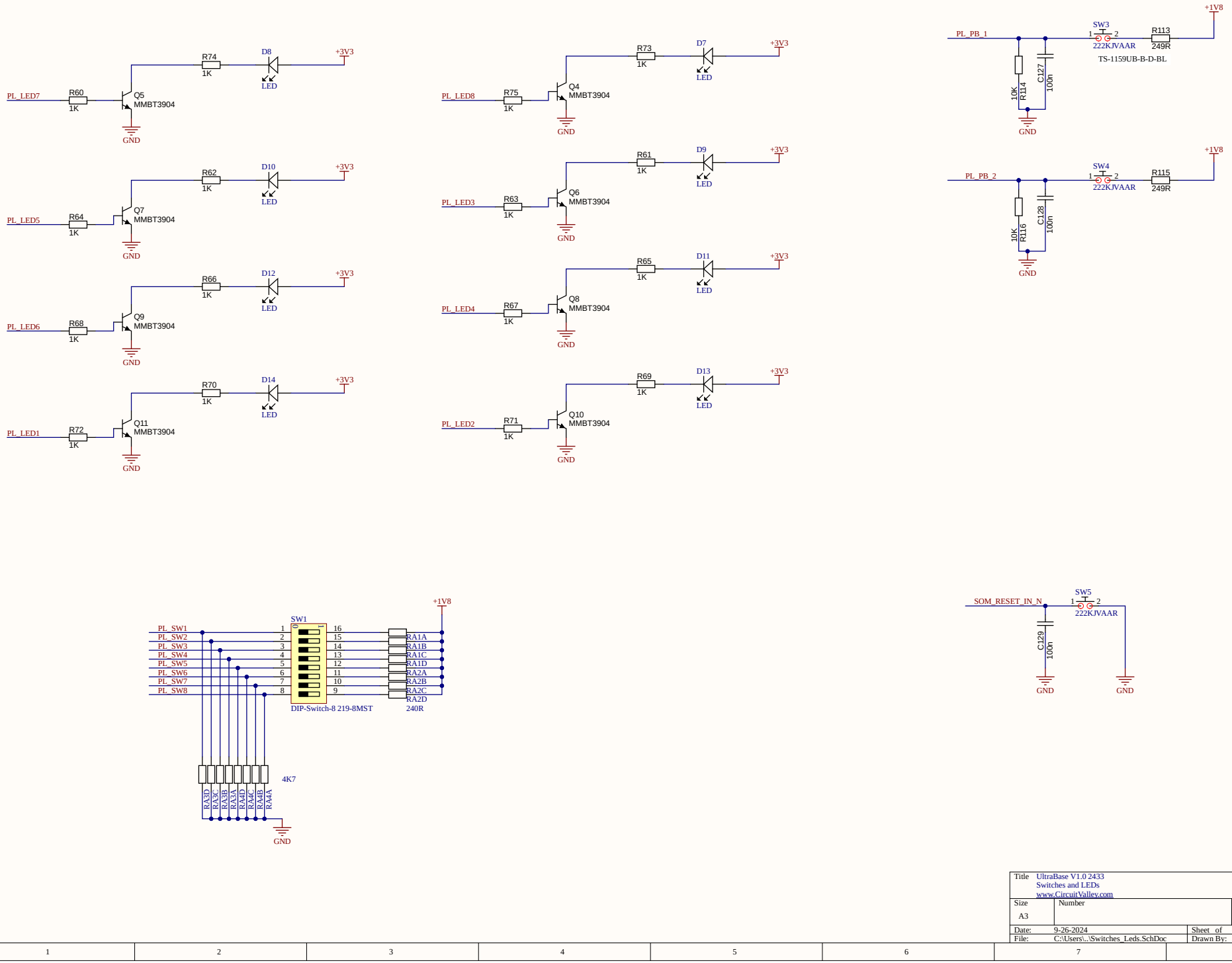
Title	UltraBase V1.0 2433 Ethernet www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\Ethernet.SchDoc		Drawn By:



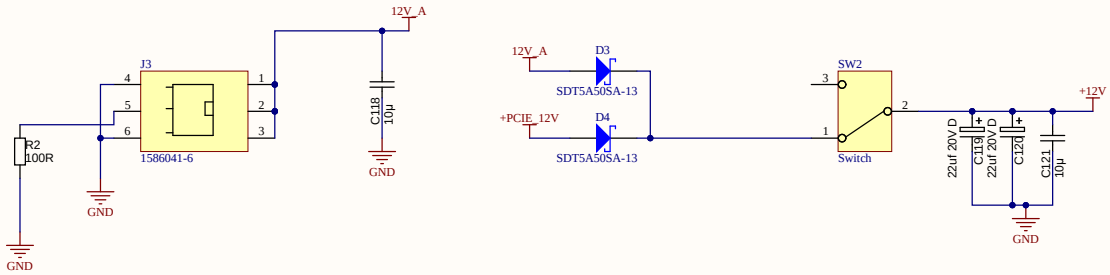
Title	UltraBase V1.0 2433 DisplayPort www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\DisplayPort.SchDoc		Drawn By:

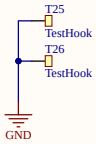


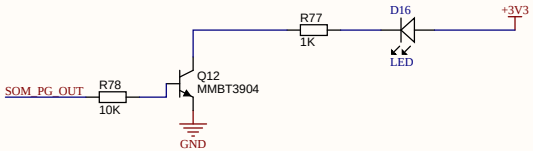
Title	UltraBase V1.0 2433 Clock Generator www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\Clock.SchDoc		Drawn By:

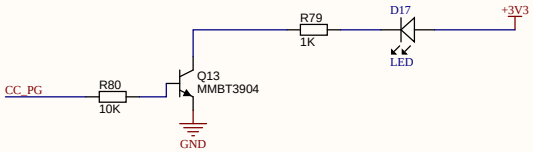


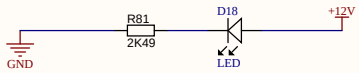
Title	UltraBase V1.0 2433 Switches and LEDs www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\Switches_Leds.SchDoc		Drawn By:

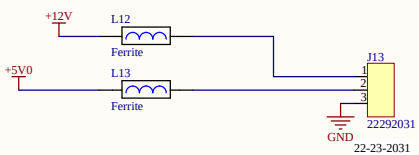


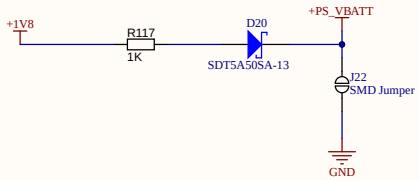


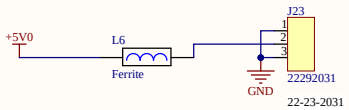




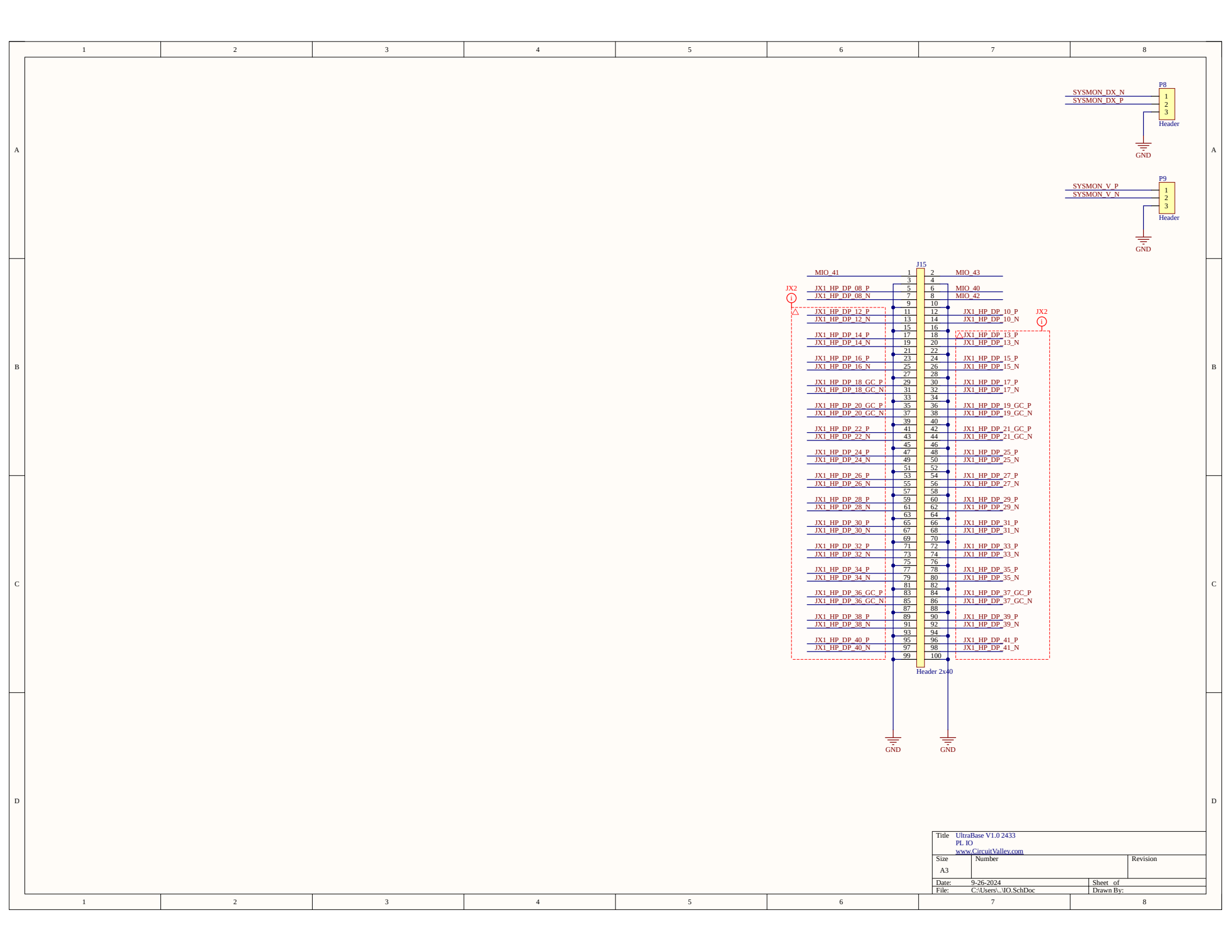


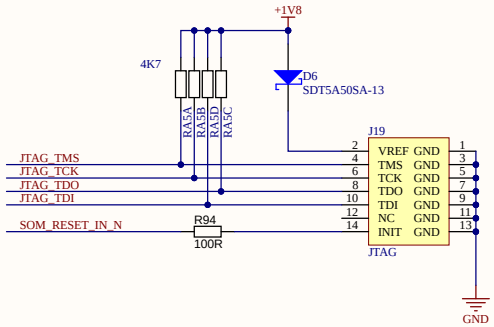




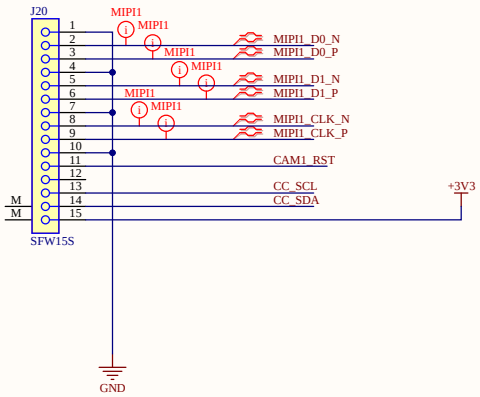


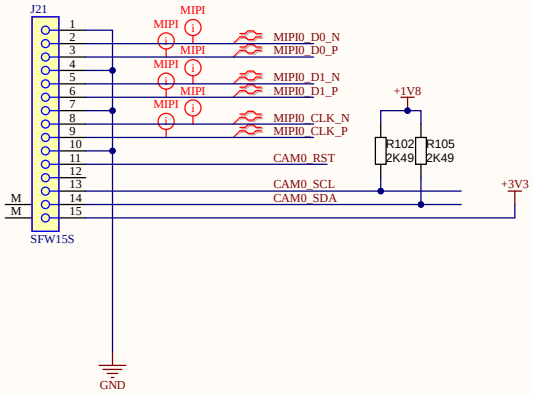
Title	UltraBase V1.0 2433 Power Input www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\PWR_INP.SchDoc		Drawn By:

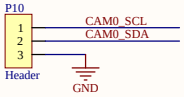

P8
SYSMON_DX_N
SYSMON_DX_P
1
2
3
Header
GND
P9
SYSMON_V_P
SYSMON_V_N
1
2
3
Header
GND
J15
JX2
JX2
MIO_41
MIO_43
JX1_HP_DP_08_P
JX1_HP_DP_08_N
MIO_40
MIO_42
JX1_HP_DP_12_P
JX1_HP_DP_12_N
JX1_HP_DP_10_P
JX1_HP_DP_10_N
JX1_HP_DP_14_P
JX1_HP_DP_14_N
JX1_HP_DP_13_P
JX1_HP_DP_13_N
JX1_HP_DP_16_P
JX1_HP_DP_16_N
JX1_HP_DP_15_P
JX1_HP_DP_15_N
JX1_HP_DP_18_GC_P
JX1_HP_DP_18_GC_N
JX1_HP_DP_17_P
JX1_HP_DP_17_N
JX1_HP_DP_20_GC_P
JX1_HP_DP_20_GC_N
JX1_HP_DP_19_GC_P
JX1_HP_DP_19_GC_N
JX1_HP_DP_22_P
JX1_HP_DP_22_N
JX1_HP_DP_21_GC_P
JX1_HP_DP_21_GC_N
JX1_HP_DP_24_P
JX1_HP_DP_24_N
JX1_HP_DP_25_P
JX1_HP_DP_25_N
JX1_HP_DP_26_P
JX1_HP_DP_26_N
JX1_HP_DP_27_P
JX1_HP_DP_27_N
JX1_HP_DP_28_P
JX1_HP_DP_28_N
JX1_HP_DP_29_P
JX1_HP_DP_29_N
JX1_HP_DP_30_P
JX1_HP_DP_30_N
JX1_HP_DP_31_P
JX1_HP_DP_31_N
JX1_HP_DP_32_P
JX1_HP_DP_32_N
JX1_HP_DP_33_P
JX1_HP_DP_33_N
JX1_HP_DP_34_P
JX1_HP_DP_34_N
JX1_HP_DP_35_P
JX1_HP_DP_35_N
JX1_HP_DP_36_GC_P
JX1_HP_DP_36_GC_N
JX1_HP_DP_37_GC_P
JX1_HP_DP_37_GC_N
JX1_HP_DP_38_P
JX1_HP_DP_38_N
JX1_HP_DP_39_P
JX1_HP_DP_39_N
JX1_HP_DP_40_P
JX1_HP_DP_40_N
JX1_HP_DP_41_P
JX1_HP_DP_41_N
Header 2x40
GND
GND
Title UltraBase V1.0 2433
PL IO
www.CircuitValley.com
Size
A3
Number
Revision
Date: 9-26-2024
Sheet of
File: C:\Users\..\IO.SchDoc
Drawn By:

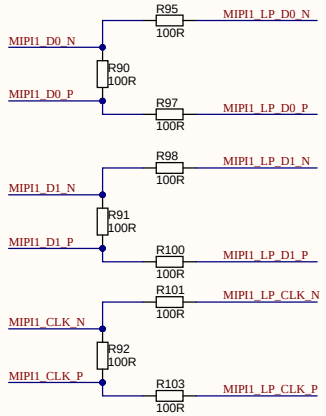


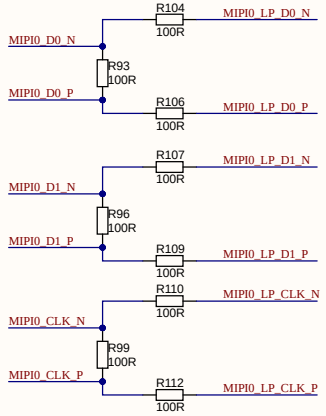
Title	UltraBase V1.0 2433 JTAG www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\JTAG.SchDoc		Drawn By:



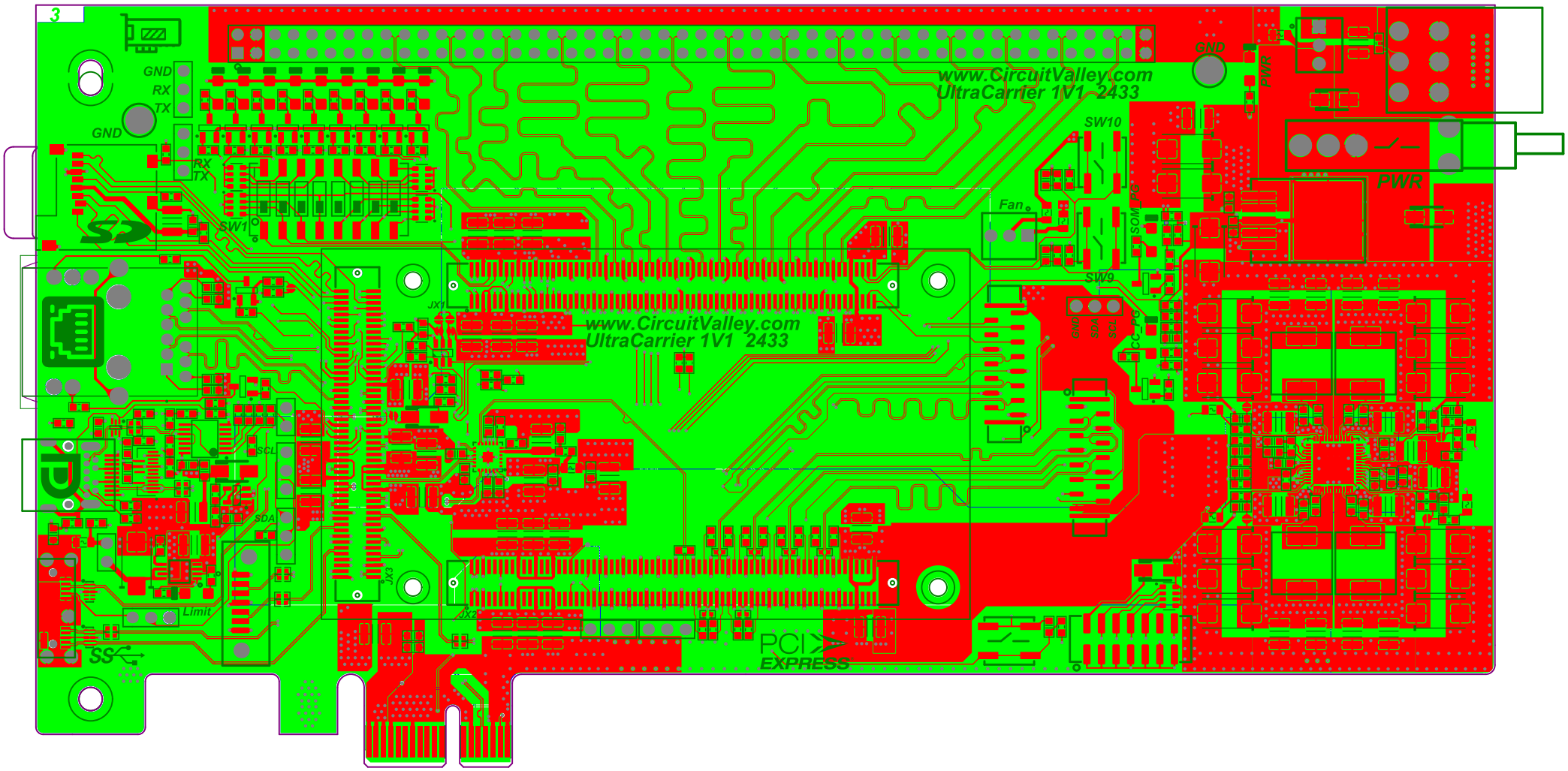








Title	UltraBase V1.0 2433 MIPI www.CircuitValley.com	
Size A3	Number	Revision
Date: 9-26-2024		Sheet of
File: C:\Users\..\MIPI.SchDoc		Drawn By:


3
GND
RX
TX
GND
RX
TX
SW1
www.CircuitValley.com
UltraCarrier 1V1 2433
SW10
PWR
Fan
SW9
JX1
www.CircuitValley.com
UltraCarrier 1V1 2433
SCL
SDA
JX3
JX2
Limit
PCI
EXPRESS

Comment	Description	Designator	Footprint	LibRef	Quantity	Value
10µ	Capacitor SMD Cer 10uF 20% 16V X5R 1210	C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C27, C35, C48, C60, C62, C70, C75, C76, C77, C79, C83, C84, C95, C97, C99, C109, C110, C111, C112, C116, C118, C121, C124, C125	CAPC3225N	CAPC_10u_1210_50V	45	10µ
100n	Capacitor SMD Cer 120pF 10% 50V C0G 0603, Capacitor SMD Cer 1nF 10% 100V X7R 0603	C22, C24, C29, C31, C34, C40, C41, C42, C43, C49, C53, C54, C64, C67, C68, C69, C89, C90, C91, C92, C93, C96, C98, C100, C123, C126, C127, C128, C129	CAPC1608N	CAPC_120p_0603_C0G_50V, CAPC_1n_0603_100V	29	100n
100n	Capacitor SMD Cer 100nF 10% 16V X5R 0402	C25, C26, C36, C37, C38, C39, C44, C45, C46, C47, C85, C86	CAPC1005N	CAPC_100n_0402_16V_10%	12	100n
47uf 20V D	Capacitor SMD Tant 330uF 20% 6V3 2917	C83, C61, C71, C72, C73, C74, C79, C80, C81, C82, C101, C102, C103, C104, C119, C120	CAPMP7343X310N	CAPT_330u_2917_6V3	16	22uf 20V D
4u7	Capacitor SMD Cer 120pF 10% 50V C0G 0603	C50, C51, C52	CAPC1608N	CAPC_120p_0603_C0G_50V	3	4u7
10n	Capacitor SMD Cer 120pF 10% 50V C0G 0603	C55, C56, C57, C58, C59, C85, C86, C87, C88	CAPC1608N	CAPC_120p_0603_C0G_50V	9	10n
1u	Capacitor SMD Cer 120pF 10% 50V C0G 0603	C63, C94, C105, C106, C107, C108, C113, C114, C115	CAPC1608N	CAPC_120p_0603_C0G_50V	9	1u
TPD8S009DSMR	Display Port ESD	D1	WSON-15	TPD8S009DSMR	1	
TPD4E001DRLR	ESD Protection	D2	SOT5x3	TPD4E001DRLR	1	
SDT5A5059A-13	Diode	D3, D4, D5, D6, D20	DO-214AC	Diode	5	
LED	LED	D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19	1206_LED	LED	13	
PCIe_x1	PCIe X1 Card edge	J1	PCIe_x1	PCIe_x1	1	
GTFP08441BEU	MicroSD card	J2	GTFP08441BEU	GTFP08441BEU	1	
1586041-6	PCIE Molex Power	J3	1586043-6	1586041-6	1	
SAT3M3232042TR	Sata Connector Vertical	J5	SAT3M3232042TR	SAT3M3232042TR	1	
L829-1J1T-43	Maglack 1000BASE-T | 1X1 | 3w-OG/Y | Tab Up | Low Profile | RoHS	J6	L829-1J1T-43	L829-1J1T-43	1	
GSB343T33HR	USB3.2 Micro B	J7	GSB343T33HR	GSB343T33HR	1	
G46M2023201AWEU	Mini Display Port	J8	G46M2023201AWEU	G46M2023201AWEU	1	
22292031	22292031	J13, J23	22-23-2031	22292031	2	
Header 2x40		J15	Header_2x50	Header_2x50	1	
JTAG	JTAG	J19	878321420	Xilinx_JTAG	1	
SFW15S	15 pin 1mm FPC, RPI Camera	J20, J21	SFW15S-2STMAE1LF	SFW15S	2	
Female/Receptcale	140 Pin Ultrascale SOM Connector with ultrazed specific SOM net delay	JX1, JX2	5177985-6	5177985-6_Ultrazed_JX1, 5177985-6_Ultrazed_JX2	2	
Female/Receptcale	100 Pin Ultrascale SOM Connector with ultrazed specific SOM net delay	JX3	5177985-4	5177985-4_Ultrazed	1	
3.3uh		L1, L2, L3	1217AS-H-3R3N=P3	inductor_smd	3	
1.5uh		L4	1217AS-H-3R3N=P3	inductor_smd	1	
Ferrite	Ferrite Bead	L5, L6, L7, L8, L9, L10, L11, L12, L13	FBC2012N	Ferrite Bead	9	
Ultrazed Mechanical		M1	ULTRAZED_SOM_Mechanical	UltrazedSOMMechanical	1	
Header	2 Pin Header	P1, P4	Header2x	Header_2Pin	2	
Header	3 Pin Header	P2, P3, P5, P6, P7, P8, P9, P10	Header3x	Header_3Pin	8	
NPN_MMBT3904	Transistor SMD NPN 40V 0.2A 350mW SOT-23	Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q13	SOT23-3	NPN_MMBT3904	13	
RES_NL_0603	Resistor, SMD chip	R1, R42, R43, R47, R49, R52	RESC1608N	RES_NL_0603	6	649R
RES_NL_0603	Resistor, SMD chip	R2, R57, R94, R95, R97, R98, R100, R101, R103, R104, R106, R107, R109, R110, R112, R119	RESC1608N	RES_NL_0603	16	100R
RES_NL_0603	Resistor, SMD chip	R3	RESC1608N	RES_NL_0603	1	20K
RES_NL_0603	Resistor, SMD chip	R4	RESC1608N	RES_NL_0603	1	12K
RES_NL_0603	Resistor, SMD chip	R5, R6, R8, R10, R11, R17, R23, R27, R29, R55, R58, R59, R78, R80, R82, R83, R84, R114, R116	RESC1608N	RES_NL_0603	19	10K
RES_NL_0603	Resistor, SMD chip	R7, R35, R37	RESC1608N	RES_NL_0603	3	49R9
RES_NL_0603	Resistor, SMD chip	R9, R16, R21, R26, R33, R34, R38, R41	RESC1608N	RES_NL_0603	8	0R
RES_NL_0603	Resistor, SMD chip	R12, R44, R45, R46, R50, R54	RESC1608N	RES_NL_0603	6	10R
RES_NL_0603	Resistor, SMD chip	R13, R14, R28	RESC1608N	RES_NL_0603	3	1M
RES_NL_0603	Resistor, SMD chip	R15, R36, R39	RESC1608N	RES_NL_0603	3	100K
RES_NL_0603	Resistor, SMD chip	R18, R53, R108, R111, R113, R115, R118	RESC1608N	RES_NL_0603	7	249R
RES_NL_0603	Resistor, SMD chip	R19	RESC1608N	RES_NL_0603	1	191R
RES_NL_0603	Resistor, SMD chip	R20, R40, R56, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R77, R79, R87, R88, R89, R117	RESC1608N	RES_NL_0603	25	1K
RES_NL_0603	Resistor, SMD chip	R22, R76	RESC1608N	RES_NL_0603	2	2R
RES_NL_0603	Resistor, SMD chip	R24, R30	RESC1608N	RES_NL_0603	2	1.78K
RES_NL_0603	Resistor, SMD chip	R25	RESC1608N	RES_NL_0603	1	10K NC
RES_NL_0603	Resistor, SMD chip	R31, R81, R102, R105	RESC1608N	RES_NL_0603	4	2K49
RES_NL_0603	Resistor, SMD chip	R32	RESC1608N	RES_NL_0603	1	1K5
RES_NL_0603	Resistor, SMD chip	R48, R51	RESC1608N	RES_NL_0603	2	10R NC
RES_NL_0603	Resistor, SMD chip	R85	RESC1608N	RES_NL_0603	1	47K
RES_NL_0603	Resistor, SMD chip	R86	RESC1608N	RES_NL_0603	1	15K
Thick Film	Resistor SMD chip 3.3k Ohm 0.063W 5% 0402	R90, R91, R92, R93, R96, R99	RESC1005N	RES_3K3_0402_5%	6	100R
Thick Film	Resistor array 4-pack 5% 0402 SMD	RA1, RA2	YC164	RES_ARRAY_0402_5%	2	240R
Thick Film	Resistor array 4-pack 5% 0402 SMD	RA3, RA4, RA5	YC164	RES_ARRAY_0402_5%	3	4.7K
DIP-Switch-8 219-8MST	DIP Switch 8 Position	SW1	219-8MST	DIP-Switch-8	1	
Switch	Switch SPDT	SW2	1MS1T1B4M7QES	Switch_SPDT	1	
222KJVAAR		SW3, SW4, SW5	222KJVAAR	PushButton	3	
TestHook		T25, T26	1502-x	TestHook	2	
MIC29302WU		U1	TO-263-5	MIC29302WU	1	
TPD4E05U06DQAR	ESD Protection	U2, U4	uDFN-10	TPD4E05U06DQAR	2	
MIC2544-1YMM	High Side Switch, Current limit programmable	U3	MSOP8	MIC2544	1	
IRPS5401M	PMIC Flexible Power Management Unit	U5	INF-PG-QFN-56-7X7	IRPS5401M	1	
FIN1019MTCX	Differential Buffer	U6	TSSOP14 PW G14	FIN1019MTCX	1	
5P49V6975A000LTGI	Programmable Clock Generators	U7	LGA24	5P49V6975	1	
AP7365-WG-7	LDO Reg ADJ 600mA	U8	SOT23-5	AP7365-WG-7	1	
24AA025E	MAC EEROM	U9	SOT23-6	24AA025E	1	